Learning Objective 2.2: I can simulate a dipole antenna using an EM simulation tool and interpret the results against analytical predictions.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Segmentation arithmetic.** You are about to model a straight center-fed dipole at $f = 2.45$GHz in NEC. The wire is 0.8 mm in diameter, so the radius is $a = 0.4$ mm. Before touching the keyboard, size the model on paper.

   (a) Find the free-space wavelength and the physical length of a half-wave dipole at this frequency.

   $\lambda =$ ☐

   $\lambda/2 =$ ☐

   (b) You choose $N = 15$ segments. Find the segment length Δ and show that it satisfies both NEC bounds: $\Delta < \lambda/20$ and $\Delta > 8a$.

   $\Delta =$ ☐

   (c) What is the largest *odd* segment count this wire will tolerate before it violates the $\Delta > 8a$ rule?

   $N_{max} =$ ☐

   (d) In one or two sentences, explain why the segment count must be odd.

2. **Predict, then account for the difference.** You model the 2.45 GHz dipole of Question 1 at *exactly* $\lambda/2 = 61.2$ mm, with 15 segments and a 1 V source on the center segment. NEC returns $Z_{in} = 86.9 + j46.2\Omega$ and a peak gain of 2.17 dBi.

   (a) State the four predictions Lesson 7 makes for a half-wave dipole: input impedance at exactly $\lambda/2$, resonant length, input resistance at resonance, and directivity in dBi.

   (b) Compute the percent difference between the simulated and predicted values of R_{in} and of X_{in}.

   R_{in} error =

   X_{in} error =

   (c) In two or three sentences, account for the resistance difference. Be specific about the mechanism — "simulation error" earns nothing.

   (d) Trimming brings this wire to resonance at 0.473 λ. Find the resonant length in millimetres and the percent shortening from $\lambda/2$.

   L_{res} =

   trim =

3. **Reading a convergence study.** A classmate models a dipole of length exactly 0.5 λ with wire radius $a = 0.001\ \lambda$ and re-runs it at five segment counts:

Segments N	R_{in} (Ω)	X_{in} (Ω)
5	79.9	+35.9
11	81.9	+43.9
21	83.4	+45.6
41	84.5	+46.5
81	85.4	+47.2

(a) Compute the percent change in R_{in} from each row to the next.

last step =

(b) Find the largest segment count the $\Delta > 8a$ rule permits for this wire. Which rows of the table violate it?

N_{max} =

(c) Which segment count would you defend for this model, and why? Your answer must reference both the convergence trend and the kernel limit.

(d) Explain in one or two sentences why gain converges at a lower segment count than input impedance does.

4. **The average-gain audit.** After a full-sphere pattern run of a lossless dipole in free space, NEC reports an average power gain of 1.15.

   (a) What does an average power gain of exactly 1.000 tell you about a lossless free-space model, and why is that a test rather than a result?

   (b) Express the reported 1.15 in decibels, and state whether you would accept the run.

   avg gain = $\boxed{}$

   (c) Give two plausible causes of an average gain this far from 1.000, and the specific check that would confirm each.

5. **Why the current comes first.** A classmate claims that NEC "computes the radiation pattern directly from the geometry."

   (a) In two or three sentences, explain why a method-of-moments code must solve for the current distribution before it can produce a pattern. Refer to the radiation integral.

   (b) The solver drives the center segment with 1 V and reports a feed current of $8.90 - j4.80$ mA. Find Z_{in}, then the magnitude of the reflection coefficient and the VSWR referred to 50Ω.

   $Z_{in} =$ ☐

   $|\Gamma| =$ ☐

   VSWR = ☐

   (c) You need this antenna matched to 50Ω. Would you trim the wire to resonance or add a matching network? Give the trade-off in one or two sentences.

Documentation: ____________________